Qn. Set Code-1

Semester: 4th
Programme: B.Tech
Branch: CSSE

SPRING END SEMESTER EXAMINATION-2024

4th Semester B.Tech

PROBABILITY AND STATISTICS

MA 2011

(For 2021 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any SIX questions.
Question paper consists of four SECTIONS i.e. A, B, C and D.
Section A is compulsory.
Attempt minimum one question each from Sections B, C, D.
The figures in the margin indicate full marks.
All parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions: [1 × 10]

(a) Find the value of k, if the probability function of the random variable X is defined by $f(x) = k\pi^{-x}, x = 1,2,\cdots$.

(b) For any events A and B with $P(B) > 0$, then prove that $P(A'|B) < P(A')$ if $P(A|B) > P(A)$.

(c) If $A = \{101, 110\}, B = \{011, 110\}, C = \{011, 101\}$, then find $P(A \cup B|C)$

(d) Compute sample variance of the data 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55

(e) What is the mean and variance of the random variable $Z = \frac{X-\mu}{\sigma}$ where X is exponentially distributed random variable?

(f) Find the value of k, if the joint probability density function is $f(x, y) = k(x^2 + y^2)$, $20 \leq x \leq 30$, $20 \leq y \leq 30$ and 0 otherwise.

(g) Find the standard deviation for Binomial distribution is 1. If the probability of success is $p = 0.5$, then find the number of trials n.

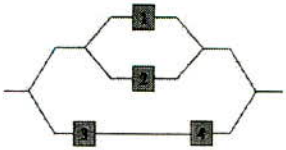
(h) If the sample points 2, 4, 5, 6, 8 satisfy the Binomial distribution with parameter with $n = 20$, find the value of the parameter p.

(i) If $p(1,1) = p(2,2) = p(3,3) = 0, p(2,1) = p(1,2) = p(3,1) = p(1,3) = p(3,2) = p(2,3) = \frac{1}{6}$, then find $P(X \leq 2 \text{ and } Y \leq 1)$.

(j) If X and Y are independent random variables with standard deviations $\sigma_X = \frac{1}{3}$ and $\sigma_Y = \frac{1}{4}$ respectively, then find $V(3X - 4Y)$.

SECTION-B

2. (a) Consider the system of components connected as in the accompanying picture. [4]



Components 1 and 2 are connected in parallel, so that subsystem works iff either 1 or 2 works; since 3 and 4 are connected in series, that subsystem works iff both 3 and 4 work. If components work independently of one another and P(component works)=0.8. Calculate P(system works) and P(only component 1 works).

(b) Let A, B, C be the events that a system has zero, minor and major defect, respectively and $P(A) = 0.12$, $P(B) = 0.07$, $P(C) = 0.05$, $P(A \cup B) = 0.13, P(B \cup C) = 0.1$, $P(A \cup C) = 0.14, P(A \cap B \cap C) = 0.01$. [4]

(i) Find the probability of the system has either zero, minor or major defect.

(ii) Find the probability that the system has only minor defects.

3. (a) A consumer organization that evaluates new automobile customarily reports the number of major defects in each car examined. Let X denote the number of major defects in a randomly selected car of a certain type. The cdf of X is as follows: [4]

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.06 & 0 \le x < 1 \\ 0.19 & 1 \le x < 2 \\ 0.39 & 2 \le x < 3 \\ 0.67 & 3 \le x \le 4 \\ 0.92 & 4 \le x < 5 \\ 0.95 & 5 \le x < 6. \\ 1 & 6 \le x \end{cases}$$

Calculate (a) $P(2 \le X \le 5)$ (b) $V(X)$

(b) Find the $E(X)$ and $E(X-\mu)^2$ of the probability function exponential distribution related with the parameter $\lambda > 0$ where μ is the mean of X. [4]

SECTION-C

4. (a) Find the mean and variance of discrete uniform random variable X where $X = \{1,2,3,\cdots,n\}$. [4]

(b) In a commuting to work, a professor must first get on a bus near her house and then transfer to a second bus. If the waiting time (in minutes) at each stop has uniform distribution with $A = 0$ and $B = 5$, then it can be shown that the total waiting time Y has the pdf [4]

$$f(y) = \begin{cases} \frac{y}{25} & 0 \le y < 5 \\ \frac{2}{5} - \frac{y}{25} & 5 \le y \le 10 \\ 0 & \text{otherwise.} \end{cases}$$

a. Compute and sketch the cdf of Y.

b. Find 95th percentile of $X = 2Y + 1$.

5. (a) Airlines sometimes overbook flights. Suppose that for a plane with 50 seats, 55 passengers have tickets. Define the random variable Y as the number of ticketed passengers who actually shows up for the flight. The probability mass function of Y appears in the accompanying table. [4]

y	45	46	47	48	49	50	51	52	53	54	55
$p(y)$	0.05	0.09	0.12	0.15	0.24	0.17	0.07	0.05	0.03	0.02	0.01

(i) If you are the third person on the standby list, what is the probability that you will be able to take the flight?

(ii) Determine the probability that Y is within 1 standard deviation of its mean value.

(b) If $X \sim N(\mu, \sigma^2)$ with probability distribution function $F_X(x)$ and $Z = \frac{X-\mu}{\sigma}$ with probability distribution function $\Phi(z)$, then prove that [4]

(i) $F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$

(ii) $F_X(\mu) = 0.5$.

6. (a) The joint exponential density function is [4]

$$f(x,y) = \begin{cases} \lambda_1\lambda_2 e^{-(\lambda_1 x+\lambda_2 y)}; & x > 0, y > 0 \\ 0; & \text{otherwise.} \end{cases}$$

(i) Verify the random variables X and Y are independent.

(ii) Compute the value of $Corr(X,Y)$.

(b) Compute $V(X)$, $V(Y)$ if the joint probability function of X and Y is [4]

$$f(x,y) = \begin{cases} x+y, (0 \le x \le 1; 0 \le y \le 1) \\ 0, \quad (\text{otherwise}) \end{cases}$$

SECTION-D

7. (a) A service station has both self-service and full-service islands. On each island, there is a single regular unleaded pump with two hoses. Let X denote the number of hoses being used on the self-service island at a particular time, and let Y denote the number of hoses on the full-service island in use at that time. The joint pmf of X and Y appears in the accompanying tabulation. [1+3]

		y		
$p(x,y)$		0	1	2
x	0	0.10	0.04	0.02
	1	0.08	0.20	0.06
	2	0.06	0.14	0.30

a. Compute $P(X \leq 1 \text{ and } Y \leq 1)$.

b. Compute $P(X \leq 1 \text{ or } Y \leq 1)$

(b) Find a 99% confidence interval for the mean μ of a normal population with variance 4 from the sample 30, 42, 40, 34, 48, 50, 31, 43, 41, 35 where $P(Z \leq -2.576) = 0.005$. [4]

8. (a) Let X be a normal random variable with variance $\sigma^2 = 15$. Using a sample size $n = 20$ with mean $\bar{x}$, test the hypothesis $\mu = \mu_0 = 24$ against the alternative $\mu \neq \mu_0$ with significance level $\alpha = 5\%$ where $P(Z \leq -1.960) = 0.025$. [4]

(b) Find the maximum likelihood estimate of θ in the probability mass function [4]

$$f(x) = \begin{cases} \frac{\theta^x}{x!} e^{-\theta}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

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